

**ANALYZING MACHINE LEARNING AND DEEP LEARNING
TECHNIQUES FOR BRAIN TUMOR CLASSIFICATION
ACROSS MRI, CT AND PET**

Abstract

The chapter presents the research topic on the analysis and comparison of machine learning and deep learning algorithms to classify brain tumours using **MRI**, **CT** and **PET** images. It underscores the importance of automated tumour detection as a method of enhancing both clinical efficacy and diagnostic quality and overcoming the drawbacks of a single-modality and small-scale research. The chapter sets the purpose of the research, objectives, and research design, creating a work outline on how to test both convolutional neural networks and support vector machines through quantitative analysis to benefit the comprehensive classification of medical images.

Table of Contents

CHAPTER 1: INTRODUCTION	4
1.1 Chapter introduction.....	4
1.2 Motivation and background of the research.....	4
1.3 Problem statement	6
1.4 Aim, objectives and research question.....	7
Project aim	7
Research objectives	7
Research question	8
1.5 Summary of the project plan	8
1.6 Adapted approach.....	9
1.7 Dissertation structure.....	10
1.8 Chapter summary	11
References	12

CHAPTER 1: INTRODUCTION

1.1 Chapter introduction

The medical imaging field of automated detection of brain tumours is a fast-developing field of convergence between medical imaging, machine learning and clinical decision support. Proper automatic recognition of tumour presence and type based on imaging modalities has effects on diagnostic, treatment planning and prognosis, and could decrease diagnostic delays and inter-reader variability. The technologies of principled developments in representation learning, specifically using deep convolutional neural networks, have provided significant changes to image-based tumour detection of public magnetic resonance imaging databases (Xie et al., 2022).

Multimodal imaging, which integrates anatomical and metabolic data using MRI, CT and PET has the potential to enhance classification performance although has not been widely studied in comparative algorithmic studies. This chapter provides the motivation, problem statement, objectives and intended course to compare convolutional neural networks and support vector machines on the data and data types of MRI, CT and PET of brain tumours classification.

1.2 Motivation and background of the research

The management aspect of the brain tumours lies in the imaging process since the staging, radiotherapy feedback, surgical planning processes and the images yielded by the imaging process influence clinical decision-making; sharply produced imaging patterns can alter clinical guidelines. Large repositories of images are time-consuming and subject to human interpretation differences, thus the need to use automated, reproducible applications that can aid clinicians (Aamir et al., 2022). Convolutional neural networks in particular have shown state-of-the-art

results on mono-modal classification and segmentation of MRI images without the use of handcrafted features by learning hierarchical image representations in pixel data. Simultaneously, classical machine learning models like support vectors still perform well, when combined with strong features extraction or employed as hybrid systems, which demonstrates that the direct comparison still has scientific as well as practical worth.

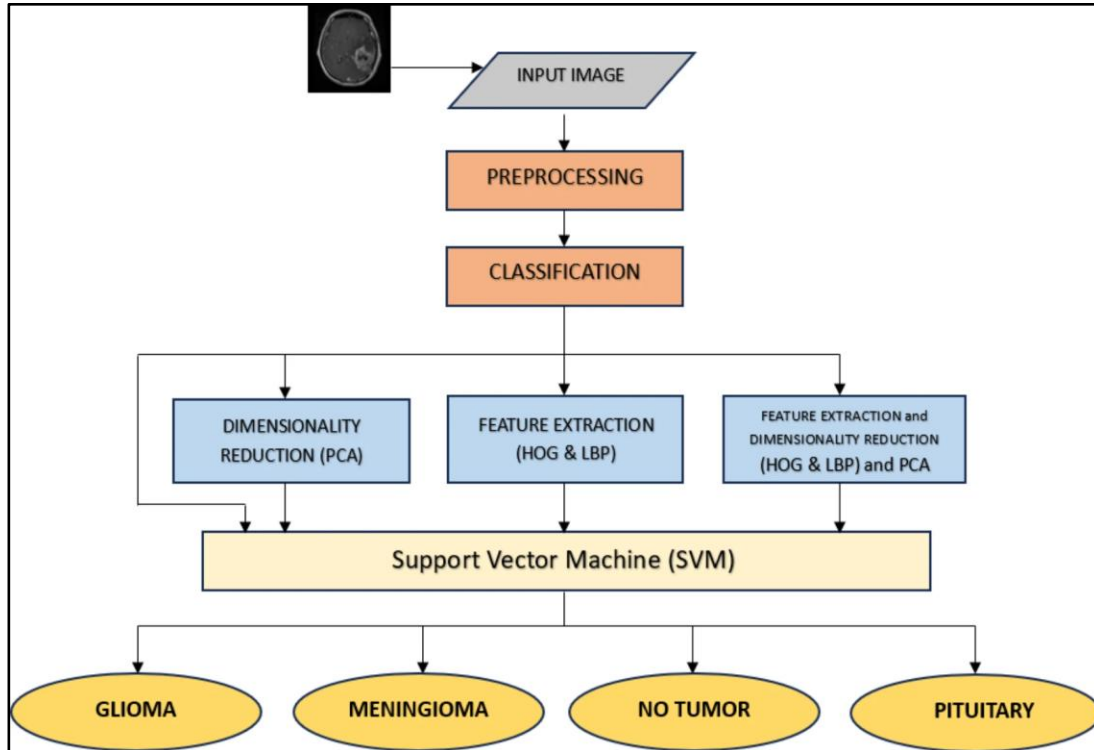


Figure 1.1: Enhancing multiclass brain tumour diagnosis using ML

(Source: Basthikodi et al., 2024)

Regardless of this improvement, the vast majority of available research studies test algorithms using one of the modules or a limited and carefully crafted dataset, creating doubts regarding their applicability to different types of imaging and acquisition settings. Multimodal strategies that combine metabolic PET data with high-resolution anatomical MRI or with CT suggest such complementary information as can enhance tumour characterisation, response assessment, and

subtype discrimination, and comparative studies benchmarking algorithms between MRI, CT and PET are rather incomplete (Basthikodi et al., 2024).

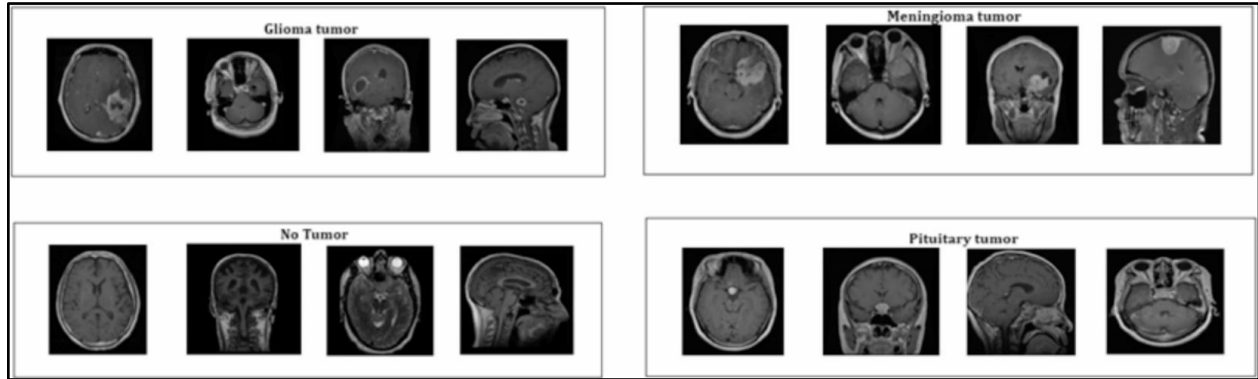


Figure 1.2: Sample MRI Images

(Source: Basthikodi et al., 2024)

The availability of datasets and the differences in their preprocessing, augmentation and assessment protocols contribute to data unavailability and difficulty in cross-study comparisons, and clinics are unintegrated by the integration of robust evidence concerning functionality, failure modes and resource necessities. The study fills the existing gaps through the creation of available publicly available MRI experiments and through the generation of a usable pipeline enabling the comparison between convolutional neural networks and support aversions of modalities, preprocessing settings and evaluation schemes (Riahi et al., 2024).

1.3 Problem statement

Although it has presented convincing demonstrations of automated tumour detection in a single-modality MRI data set, the capabilities of different families of algorithms in various imaging modalities and in aspects of acquisition variability remain uncertain. Hand-crafted tests have been found to be highly accurate on particular sets of curated images, but fail to test approaches with PET or CT images or rates of breakdown to common preprocessing decisions (Raza et al., 2023).

The lack of systematic, comparative research that compares deep learning and classical machine learning strategies in MRI, CT and PET restricts evidence-based suggestions regarding the clinical implementation (Rasheed et al., 2023). This necessitates the representation and benchmarking of representative supervised learning methods such as convolutional neural networks and support vector machines between modalities with a unified preprocessing and evaluation which benefits both modalities.

The present research is expected to recognise performance drivers, measure trade-offs between accuracy and computational/resource requirements, and suggest modality-specific guidelines to real-life brain tumour classification processes.

1.4 Aim, objectives and research question

Project aim

The objective of the current project is to conduct an efficient brain tumour classification methodology through comparing and analysing machine learning and deep learning methods across the MRI, CT, and PET scans.

Research objectives

- To investigate and preprocess the Brain MRI, CT and PET image data to develop high-quality inputs in the classification analysis.
- To implement and evaluate several supervised learning methods in the detection and classification process of brain tumours using medical images.
- To assess and benchmark the classification techniques using comprehensive performance metrics for determining the most effective approach for accurate brain tumour detection.

- To determine major attributes that affect the performance of classifications and propose recommendations towards the optimisation of brain tumour detection workflows.

Research question

What is the future of machine learning and deep learning tools in enhancing brain tumour classification processes in various imaging modalities in terms of accuracy and efficiency?

1.5 Summary of the project plan

The initial phase of the project is the acquisition and descriptive examination of the “*Brain MRI Images for Brain Tumour Detection*” dataset on Kaggle, which forms the main source of data. The preprocessing will involve image resizing, intensity normalisation, and augmentation to maintain uniform and quality inputs to be used in training the model. This project will entail the construction of two supervised learning pipelines e.g., a “*Convolutional Neural Network (CNN)*” to perform end-to-end image classification and a “*Support Vector Machine (SVM)*” to learn on extracted image features, with CNN-based image feature embeddings being used on the hybrid evaluation.

The two models would be trained, validated and evaluated strictly on cross-validation and key performance measures described as accuracy, precision, recall, F1-score and ROC-AUC to ascertain reliability on benchmarking (Asiri et al., 2024). The comparison of analysis will result in the determination of a more efficient model in modalities and preprocessing strategy, and the most efficient model should be adopted in the further evaluation and optimisation.

1.6 Adapted approach

The data-driven and adjusted strategy in this project entails an organised process of data gathering, pre-processing, model creation, and analysis.

Steps	Activities
Step 1: Dataset acquisition and inspection	● Download the Kaggle “Brain MRI Images for Brain Tumour Detection” dataset (Kaggle.com). ● Check data quality and class balance. ● Review image structure and metadata.
Step 2: Data preprocessing and preparation	● Resize and normalise images. ● Split into train, validation, and test sets.
Step 3: Feature extraction and engineering	● Extract texture and shape features for SVM. ● Use CNN layers for feature embeddings.
Step 4: Model development (CNN and SVM)	● Build a CNN using TensorFlow and Keras. ● Train SVM with RBF or linear kernel. ● Apply dropout and tuning to avoid overfitting.
Step 5: Model evaluation and benchmarking	● Measure accuracy, recall, and F1-score. ● Perform cross-validation for reliability (Mahesh et al., 2023). ● Compare CNN and SVM across modalities.

Step 6: Insights and recommendations	• Identify key performance factors. • Summarise model strengths and weaknesses. • Suggest workflow improvements for clinics.
---	--

Table 1: Adapted project approaches

This method of assessment offers a stable and transparent system to assess machine learning and deep learning models.

1.7 Dissertation structure

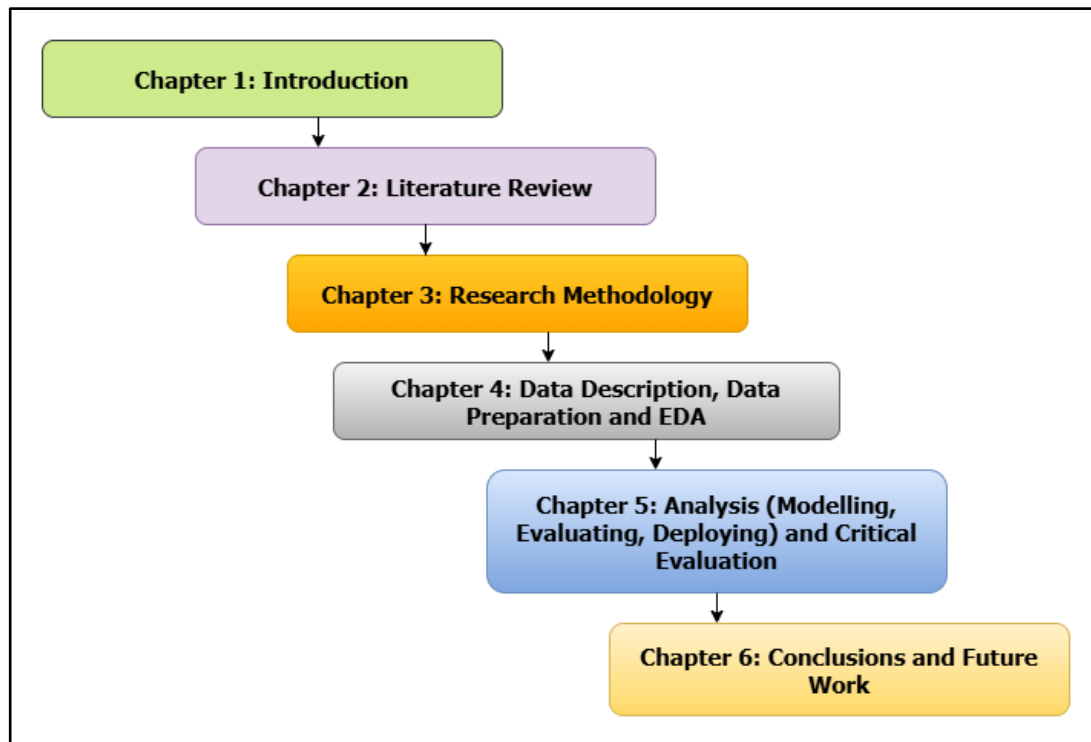


Figure 1.3: Structure of the research project

The dissertation is structured in a logical flow that builds up one after another based on theoretical underpinnings and application analysis evaluation. The sections closely follow one another, taking the reader through the background and the methodological design to the data exploration, model

implementation, critical evaluation, and final conclusions with proposals on future research directions.

1.8 Chapter summary

This chapter represented a general summary of the research project aimed at the analysis of machine learning and deep learning methods to classify brain tumours through the MRI, CT, and PET imaging modalities. It set the importance on automated diagnostic tools to enhance accuracy and efficiency of the image analysis in the field of medicine. The discussion has identified the research motivation, the background and the challenges that exist to restrict the current mono-modal techniques of brain tumour detection. The purpose and goals of the research and the research question were clearly stated to present the direction and boundaries of the project. The chapter presented the project plan in general, study methodology, and the dissertation organisation used in further chapters.

References

- Aamir, M., Rahman, Z., Dayo, Z. A., Abro, W. A., Uddin, M. I., Khan, I., ... & Hu, Z. (2022). A deep learning approach for brain tumor classification using MRI images. *Computers and Electrical Engineering*, 101, 108105.
- Asiri, A. A., Soomro, T. A., Shah, A. A., Pogrebna, G., Irfan, M., & Alqahtani, S. (2024). Optimized brain tumor detection: a dual-module approach for mri image enhancement and tumor classification. *IEEE access*, 12, 42868-42887.
<https://ieeexplore.ieee.org/iel7/6287639/10380310/10476596.pdf>
- Basthikodi, M., Chaithrashree, M., Ahamed Shafeeq, B. M., & Gurpur, A. P. (2024). Enhancing multiclass brain tumor diagnosis using SVM and innovative feature extraction techniques. *Scientific Reports*, 14(1), 26023. <https://www.nature.com/articles/s41598-024-77243-7.pdf>
- Mahesh, T. R., Geman, O., Margala, M., & Guduri, M. (2023). The stratified K-folds cross-validation and class-balancing methods with high-performance ensemble classifiers for breast cancer classification. *Healthcare Analytics*, 4, 100247.
<https://www.sciencedirect.com/science/article/pii/S2772442523001144>
- Rasheed, Z., Ma, Y. K., Ullah, I., Ghadi, Y. Y., Khan, M. Z., Khan, M. A., ... & Shehata, A. M. (2023). Brain tumor classification from MRI using image enhancement and convolutional neural network techniques. *Brain Sciences*, 13(9), 1320. <https://www.mdpi.com/2076-3425/13/9/1320>
- Raza, A., Ayub, H., Khan, J. A., Ahmad, I., S. Salama, A., Daradkeh, Y. I., ... & Hamam, H. (2022). A hybrid deep learning-based approach for brain tumor classification. *Electronics*, 11(7), 1146. <https://www.mdpi.com/2079-9292/11/7/1146>

Riahi, F., Kiani, P., Golabbakhsh, A., Khanezarrin, M., Abbaspour, M., Tabatabaei, S. A. D., ... & Mohammadi, H. (2024). Comparison of PET/CT and PET/MRI in central nervous system tumors, a narrative review. *International journal of physiology, pathophysiology and pharmacology*, 16(4), 89.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC11411248/pdf/ijppp0016-0089.pdf>

Xie, Y., Zaccagna, F., Rundo, L., Testa, C., Agati, R., Lodi, R., ... & Tonon, C. (2022). Convolutional neural network techniques for brain tumor classification (from 2015 to 2022): Review, challenges, and future perspectives. *Diagnostics*, 12(8), 1850.

<https://www.mdpi.com/2075-4418/12/8/1850>